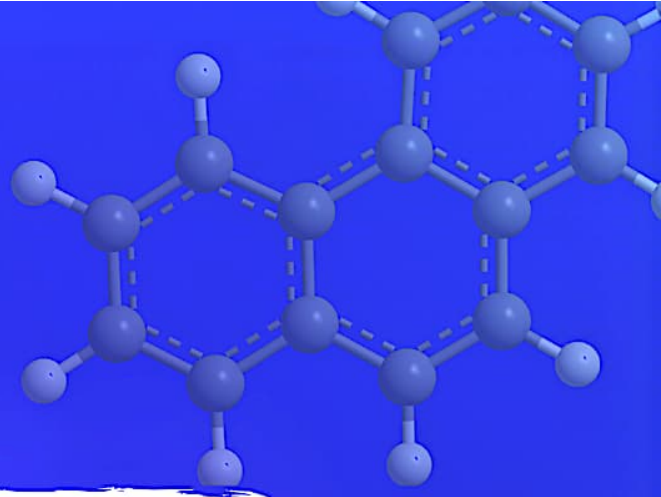




Class 9

Chapter 2

Kinematics

Chapter 2 | Class 9 | Physics

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Chapter

2

Kinematics

2.1

Scalars and Vectors

• Multiple Choice Questions (MCQs) •

- The study of motion without reference of force is:
 (A) Dynamics (B) Kinematics (C) Mechanics (D) Physics
- Which one is not a scalar quantity?
 (A) Length (B) Mass (C) Time (D) Velocity
- A vector is represented by:
 (A) Small letter (B) Capital letter
 (C) Bold face letter (D) Italic letter
- For geographical location, the reference line is:
 (A) Along x-axis (B) Along y-axis (C) Along east (D) North-South
- To find resultant vector, we apply:
 (A) Right hand rule (B) Lenz's law
 (C) Newton's law (D) Head to tail rule
- Number of vectors that can be added by head-to-tail rule:
 (A) 1 (B) 2 (C) 3 (D) Any number

• Short Questions •

- Define mechanics. Write its branches.

Ans. Mechanics: Mechanics is the branch of physics that deals with the motion of objects and the forces that change it.

Generally, mechanics is divided into two branches:

- (i) Kinematics (ii) Dynamics

- How a vector is represented symbolically?

Ans. Symbolically vector quantities are represented by bold face letter such as $\mathbf{v}$, $\mathbf{F}$ etc. Since we cannot write in bold face script on paper, so a vector is written as the letter with a small arrow over it, i.e. $\vec{v}$, $\vec{F}$, $\vec{d}$. The magnitude of a vector is given by italic letter without arrow head.

3. Differentiate between scalar and vector quantities with examples.

Ans.

Scalars	Vectors
- A scalar is that physical quantity which can be described completely by its magnitude only. - Scalar quantities can be added up like numbers. Examples: Distance, length, time, speed, energy and temperature.	- A vector is that physical quantity which needs magnitude as well as direction to describe it completely. - Vector quantities are added by head to tail rule. Examples: Displacement, velocity, acceleration, weight, force, etc.

4. How a vector is represented graphically?

Ans. A vector can be represented graphically by drawing a straight line with an arrow head at one end. The length of line represents the magnitude of the vector quantity according to a suitable scale while the direction of arrow indicates the direction of the vector.

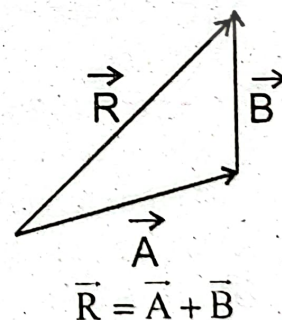


5. Define resultant vector.

Ans. **Resultant vector:** A resultant vector is a single vector that has the same effect as the combined effect of all the vectors to be added.

6. State head-to-tail rule.

Ans. **Head-to-tail rule:** To add a number of vectors, redraw their representative lines such that the head of one line coincides with the tail of the other. The resultant vector is given by a single vector which is directed from the tail of the first vector to the head of the last vector.



2.2

Rest and Motion

Multiple Choice Questions (MCQs)

7. The states of rest or motion is always:
 (A) Opposite (B) Equal (C) Relative (D) All
8. A body, changing its position is said to be:
 (A) At rest (B) In motion
 (C) has constant speed (D) has constant velocity

Short Questions

1. Define rest and motion with example.

Ans. Rest: If a body does not change its position with respect to its surroundings, it is said to be at rest.

Example: Suppose a motorcyclist is standing on the road. An observer sees that he is not changing his position with respect to his surroundings i.e., a nearby building, tree or a pole. He will say that the motorcyclist is at rest.

Motion: If a body continuously changes its position with respect to its surroundings, it is said to be in motion.

Example: When the motorcyclist is driving, he is in motion as he is changing his position with respect to his surrounding.

2. Prove that state of rest or motion is always relative.

Ans. The state of rest or motion of a body is always relative. For example, a person standing in the compartment of a moving train is at rest with respect to the other passengers in the compartment, but he is in motion with respect to an observer standing on the platform of a railway station. It shows that state of rest or motion is always relative.

2.3

Types of Motion

Multiple Choice Questions (MCQs)

9. Types of motion are:

(A) 1 (B) 2 (C) 3 (D) 4

10. The motion of a bee is:

(A) Linear (B) Random (C) Vibratory (D) Rotatory

11. To and fro motion about a fix point is called:

(A) Circular motion (B) Random motion
(C) Vibratory motion (D) Rotatory motion

12. Straight line motion is called:

(A) Linear motion (B) Random motion
(C) Rotatory motion (D) Circular motion

Short Questions

1. What is translatory motion?

Ans. Translatory Motion: If the motion of a body is such that every particle of the body moves uniformly in the same direction, it is called translatory motion.

2. Write types of translatory motion with examples.

Ans. There are three types of translatory motion.

- i. **Linear Motion:** If the body moves along a straight line, it is called linear motion.

Example: A freely falling body is the example of linear motion.

- ii. **Random Motion:** If the body moves along an irregular path, the motion is called random motion.

Example: Motion of smoke particles, motion of bee, motion of gas molecules are examples of random motion.

- iii. **Circular Motion:** The motion of a body along a circle is called circular motion.

Example: If a ball tied to one end of a string is whirled, it moves along a circle. A Ferris wheel is also an example of circular motion.

3. Define rotatory and vibratory motion with examples.

- Ans. Rotatory Motion:** If each point of a body moves around a fixed point (axis), the motion of this body is called rotatory motion.

Example: The motion of an electric fan and the drum of a washing machine dryer is rotatory motion.

Vibratory Motion: When a body repeats its to and fro motion about a fixed position, the motion is called vibratory motion.

Example: The motion of a swing in a children park is vibratory motion. The motion of see-saw is also an example of vibratory motion.

2.4

Distance and Displacement

Multiple Choice Questions (MCQs)

13. Length of actual path of motion is called:

(A) Distance (B) Displacement (C) Speed (D) Velocity

14. Displacement is a:

(A) Scalar quantity (B) Vector quantity
(C) Constant quantity (D) Variable quantity

15. SI unit of displacement is:

(A) m (B) kg (C) N (D) ms^{-1}

Short Questions

1. Differentiate between distance and displacement?

Ans.	Distance	Displacement
	- The distance is the length of actual path of the motion between two points. - Distance is a scalar quantity - The SI unit of distance is meter.	- The displacement of an object is the shortest distance between the initial and final positions of the motion. - Displacement is a vector quantity - The SI unit of displacement is meter.

2.5

Speed and Velocity

Multiple Choice Questions (MCQs)

16. Distance covered in unit time is:
 (A) Speed (B) Velocity
 (C) Acceleration (D) Average Velocity
17. The speed of sailfish is:
 (A) 70 kmh^{-1} (B) 100 kmh^{-1} (C) 110 kmh^{-1} (D) 200 kmh^{-1}
18. Velocity is a quantity:
 (A) Scalar (B) Vector (C) Constant (D) Non-physical
19. The unit of speed is:
 (A) Kg (B) ms^{-1} (C) ms^{-2} (D) cm
20. Speed of vehicles is measured by:
 (A) Micrometer (B) Galvanometer (C) Ammeter (D) Speedometer
21. Mount st. Helens erupted in:
 (A) 1980 (B) 1960 (C) 1970 (D) 1920
22. Paratrooper moves down with:
 (A) Uniform Speed (B) Constant Speed
 (C) Terminal Velocity (D) Instantaneous Velocity

Short Questions

1. Define the term speed, instantaneous speed and average speed.

Ans. **Speed:** Distance covered by a body in a unit time is called its speed.

Formula:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$v = \frac{S}{t}$$

Instantaneous speed: The speed of a vehicle that is shown by its speedometer at any instant is called instantaneous speed.

Average speed: Average speed of a body is the ratio of total distance covered by a body to total time taken to cover that distance.

Formula:

$$\text{Average speed} = \frac{\text{Total distance covered}}{\text{Total time taken}} \quad \text{or} \quad v_{av} = \frac{S}{t}$$

2. The car while moving on a circular road may have constant speed, but its velocity is changing at every instant. Why?

Ans. When a car moves on a circular road at constant speed, its velocity changes because velocity depends on both speed and direction. In circular motion, the direction continuously changes, even if the speed remains constant. This changes in direction causes the velocity to vary at every instant.

3. Define velocity and Average velocity. Write its formula and unit.

Ans. **Velocity:** The net displacement of a body in unit time is called velocity.

Formula:

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}} \quad \text{or} \quad v = \frac{d}{t}$$

Unit: The SI unit of velocity is ms^{-1} or km h^{-1} . It is a vector quantity.

Average velocity: Average velocity of a body is the ratio of total displacement covered by a body to time taken to cover that displacement.

Formula:

$$\text{Average velocity} = \frac{\text{Total Displacement}}{\text{Time}} \quad \text{or} \quad v_{\text{av}} = \frac{d}{t}$$

4. What is the difference between uniform velocity and non-uniform velocity?

Ans. The velocity is said to be uniform if the speed and direction of a moving body does not change, otherwise it is non-uniform.

2.6

Acceleration

Multiple Choice Questions (MCQs)

23. The unit of acceleration is:

(A) ms^{-2} (B) ms^{-1} (C) Km/h (D) ms

24. If rate of change of velocity is constant, then acceleration will be:

(A) Variable (B) Uniform (C) Positive (D) Negative

25. Acceleration is a quantity:

(A) Scalar (B) Vector (C) Both (A) & (B) (D) Non-physical

Short Questions

1. What is acceleration?

Ans. Acceleration: Acceleration is defined as the time rate of change of velocity. The SI unit of acceleration is ms^{-2} . Acceleration is a vector quantity. Its direction is same as that of velocity.

2. Write the formula for average acceleration.

Ans. Formula:

$$\text{Average acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

$$a_{av} = \frac{v_f - v_i}{t}$$

$$a_{av} = \frac{\Delta v}{t}$$

3. Write types of acceleration.

Ans. Positive acceleration: The acceleration is positive if the velocity of a body is increasing

Negative acceleration: Acceleration it is negative if the velocity of a body is decreasing. Negative acceleration is also called deceleration or retardation.

4. Differentiate between uniform and non-uniform acceleration.

Ans. Uniform acceleration: If time rate of change of velocity is constant, the acceleration is said to be uniform.	Non-uniform acceleration: If anyone of the magnitude or direction or both of them changes, it is called variable or non-uniform acceleration.
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2.7

Graphical Analysis of Motion

Multiple Choice Questions (MCQs)

26. It is a pictorial diagram:

- (A) symbol (B) Graph (C) Velocity (D) Acceleration

27. The point where two axes intersect each other is called:

- (A) Incentre (B) Origin (C) x-axis (D) y-axis

28. In graph, independent quantity is taken along:

- (A) y-axis (B) Any direction (C) Both axes (D) x-axis

29. When a body is moving with some acceleration, the graph will be:

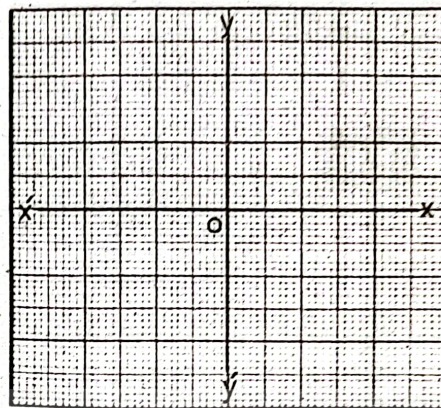
- (A) A curve (B) Straight line
(C) Parallel to x-axis (D) Parallel to y-axis

30. Horizontal line in distance-time graph indicates object is:
- (A) Moving with constant speed (B) Moving with variable speed
- (C) At rest (D) Moving constant velocity

Short Questions

1. What is meant by graph?

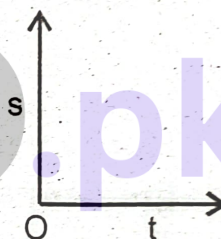
Ans. Graph: A graph is a pictorial diagram in the form of a straight line or a curve which shows the relationship between two physical quantities. In order to draw a graph, two mutually perpendicular thick lines xox' and yoy' are selected as x and y axes. The point



where the two axes intersect each other is known as origin O .

2. What is a distance-time graph?

Ans. Distance-Time Graph: Distance-time graph shows the relation between distance S and time t taken by a moving body. Time t is taken along x -axis and distance S along y -axis.



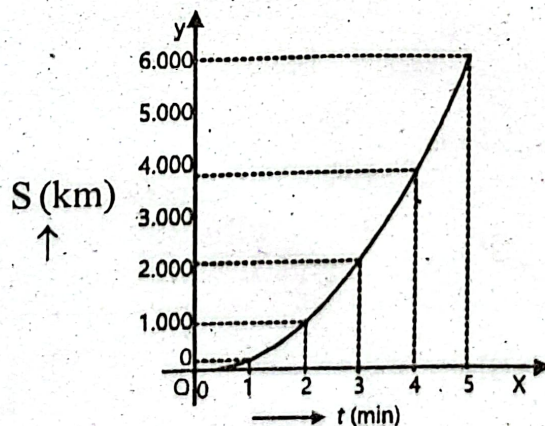
3. Explain the distance-time graph when object is at rest.

Ans. When object is at rest, graph line is horizontal parallel to x -axis.

4. Explain the distance time graph when object is moving with positive acceleration.

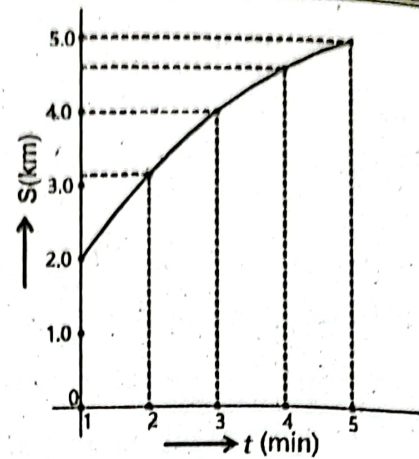
Ans. When object is moving with positive acceleration, the graph line is curved upward.

This shows the distance travelled in the same interval of time is increasing.



5. Explain the distance time graph when object is moving with negative acceleration?

Ans. When object is moving with negative acceleration, the graph line is curved downwards. This shows that distance travelled in the same interval of time is decreasing, so speed is decreasing.



2.8

Gradient of a Distance-Time Graph

Multiple Choice Questions (MCQs)

31. Gradient of distance, time graph represents:
 (A) Uniform speed (B) Average speed
 (C) Variable speed (D) Uniform velocity
32. The slope of distance-time graph is the measure of:
 (A) $\sin\theta$ (B) $\cos\theta$ (C) $\sec\theta$ (D) $\tan\theta$

Short Questions

1. How average speed of a body is measured by measuring gradient of a distance-time graph?

Ans. "Gradient of the distance-time graph is equal to the average speed of the body."

The slope or gradient of the graph is the measure of $\tan\theta$ of the triangle PQR.

So, In triangle PQR

PR = Base and represented the distance 's'

RQ = Perpendicular and represent the time interval 't'

Clearly from graph $S = S_2 - S_1$
 $t = t_2 - t_1$

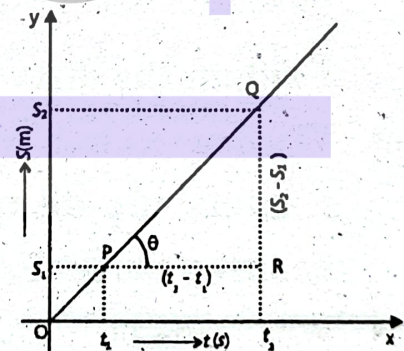
We know

$$\tan \theta = \frac{\text{perpendicular}}{\text{base}}$$

$$\text{Slope} = \frac{RQ}{PR}$$

$$\text{Slope} = \frac{S_2 - S_1}{t_2 - t_1}$$

$$\text{Slope} = \frac{S}{t}$$



2.9

Speed-Time Graph

Multiple Choice Questions (MCQs)

33. Straight line rising upward in speed-time graph represents:
 (A) Uniform speed (B) Uniform velocity
 (C) Uniform acceleration (D) Variable velocity
34. When object is moving with constant speed the speed-time graph is:
 (A) Line parallel to x-axis (B) A single point
 (C) Vertical line (D) Inclined line

Short Questions

1. Explain speed time-graph.

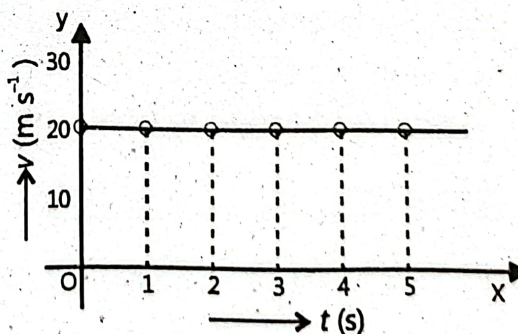
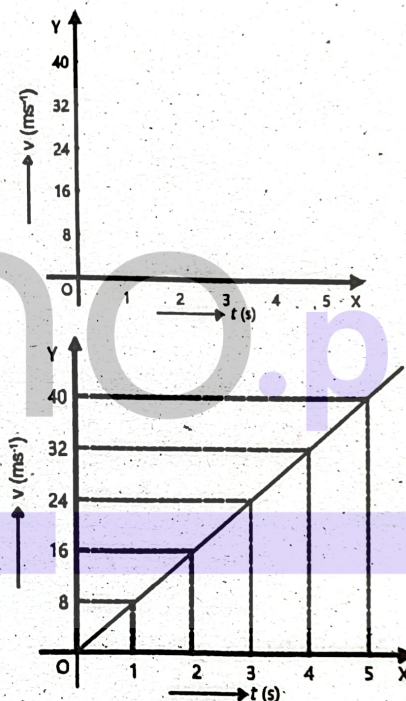
Ans. Speed time graph shows the relation between speed ' v ' and time ' t '. Time is taken along x-axis and speed (v) is taken along y-axis.

2. Explain speed time-graph when object is moving with uniformly changing speed (uniform acceleration).

Ans. When object is moving with uniformly changing speed, graph will be a straight line between time and speed axis.

3. Explain speed time-graph when object is moving with constant speed?

Ans. In this case, graph line is horizontal parallel to time x-axis. It shows that speed does not change with change in time.



2.10

Gradient of a Speed-Time Graph

Multiple Choice Questions (MCQs)

35. Gradient of speed time-graph is equal to:
 (A) Uniform speed (B) Average acceleration
 (C) Uniform velocity (D) Average velocity

36. Line parallel to time axis in speed-time graph indicates that acceleration is:

- (A) Maximum (B) Minimum (C) Zero (D) Uniform

Short Questions

1. How average acceleration of a body is measured by measuring gradient of a speed-time graph?

Ans. "Gradient of the speed-time graph is equal to the average acceleration".

Consider speed-time graph. The speed at times t_1 and t_2 is v_1 and v_2 respectively. The change in speed in time interval

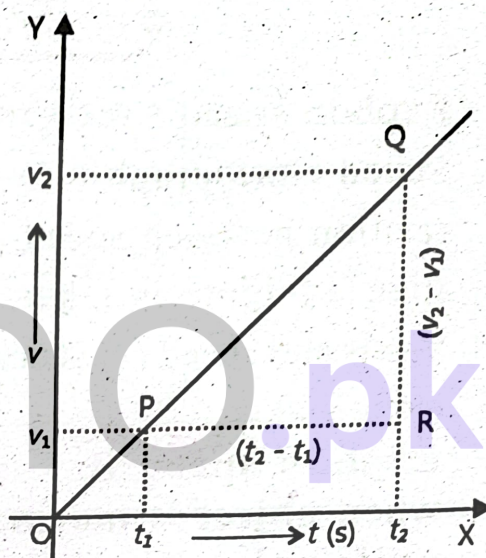
$(t_2 - t_1)$ is $(v_2 - v_1)$. Therefore,

$$\text{Slope} = \frac{\text{Change in speed}}{\text{Total time taken}}$$

$$\text{Slope} = \frac{(v_2 - v_1)}{(t_2 - t_1)}$$

$$\text{Slope} = \frac{\Delta v}{t}$$

But $\frac{\Delta v}{t} = a$, the average acceleration.



2.11

Area Under Speed-Time Graph

Multiple Choice Questions (MCQs)

37. Area under speed-time graph represents:

- (A) Distance covered by a body
(B) Displacement covered by a body
(C) Uniform acceleration
(D) Average acceleration

38. Area of triangle is given by:

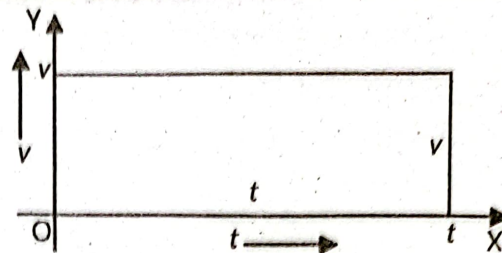
- (A) $\frac{1}{2}$ (Base $\times$ perpendicular)
(B) Length $\times$ Width
(C) Base $\times$ Perpendicular
(D) Length $\times$ Width $\times$ Height

Short Questions

1. How speed-time graph is helpful to find the distance covered by body?

Ans. For a time-interval t , the distance covered by the object is $v \times t$. The distance moved by an object can also be determined by using its speed-time graph.

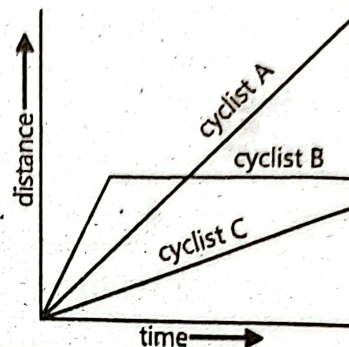
This distance can also be found by calculating the area under the speed-time graph. The area under the graph for time interval t is the area of rectangle of sides t and v . This area is equal to $v \times t$.



Mini Exercise

The distance-time graph shows the motion of three cyclists.

- What does each line on the graph represent?
- Which cyclist travelled the most distance?
- Which cyclist travelled at the greatest speed? the lowest speed? at constant speed?



- Ans. (a) Graph of line 'A' and 'C' shows that speed is uniform and body have no acceleration. Graph of line 'B' shows that speed of body is uniform and after some time object comes to rest.
- (b) Cyclist 'A' travelled most distance as area under line 'A' is most and also line 'A' is near to the distance axis (y-axis).
- (c) Cyclist 'A' travelled with greatest speed as graph is near to distance axis (y-axis). Cyclist 'C' travelled with lowest speed as graph of line 'C' is near to time axis (x-axis). Cyclist 'B' travelled at constant speed as graph of line 'B' is parallel to time axis (x-axis).

2.12

Solving Problems for Motion Under Gravity

Multiple Choice Questions (MCQs)

- There are _____ equations of motion.
 (A) 2 (B) 3 (C) 4 (D) 5
- First equation of motion is independent of:
 (A) Time (B) Acceleration (C) Distance (D) Initial speed

Short Questions

- Write equations of motion.

Ans. The equations of motion are:

$$v_f = v_i + at \quad \dots\dots\dots (1)$$

$$S = v_i t + \frac{1}{2} at^2 \quad \dots\dots\dots (2)$$

$$2aS = v_f^2 - v_i^2 \quad \dots\dots\dots (3)$$

2. What assumptions should be followed to apply equation of motion?

Ans. While applying these equations, the following assumptions are made:

- (i) Motion is always considered along a straight line.
- (ii) Only the magnitudes of vector quantities are used.
- (iii) Acceleration is assumed to be uniform.
- (iv) The direction of initial velocity is taken as positive. Other quantities which are in the direction of initial velocity are taken as positive. The quantities in the direction opposite to the initial velocity are taken as negative.

2.13

Free Fall Acceleration

Multiple Choice Questions (MCQs)

41. The value of 'g' is:

- (A) 10ms^{-1} (B) 10ms^{-2} (C) 10ms^{-3} (D) 10Ns^{-1}

42. The direction of gravitational acceleration is always:

- (A) Upward (B) Downward (C) Towards left (D) None

43. Final velocity ' V_f ' at the highest point is:

- (A) Constant (B) 10ms^{-2} (C) 10ms^{-1} (D) 0

44. Theory of relativity was proposed by:

- (A) Newton (B) Einstein (C) Faraday (D) Fleming

45. The speed of light is:

- (A) $3 \times 10^8 \text{ms}^{-1}$ (B) $3 \times 10^{-8} \text{ms}^{-1}$ (C) $3 \times 10^6 \text{ms}^{-1}$ (D) $3 \times 10^{-9} \text{ms}^{-1}$

Short Questions

1. Write the equations of motion under action of gravity.

Ans. When a body is moving under the force of gravity the acceleration "a" is replaced by "g". Thus, equations of motion for freely falling bodies can be written as:

$$v_f = v_i + gt \quad \dots\dots\dots (1)$$

$$S = v_i t + \frac{1}{2} gt^2 \quad \dots\dots\dots (2)$$

$$2gS = v_f^2 - v_i^2 \quad \dots\dots\dots (3)$$

2. What is free fall acceleration or gravitational acceleration? Write its value.

Ans. **Gravitational acceleration:** When a body is falling freely under the action of gravity of the Earth, the acceleration acting on it is the gravitational acceleration and is denoted by g. The direction of gravitational acceleration is always downwards.

Value: The value of g is 10ms^{-2} .

